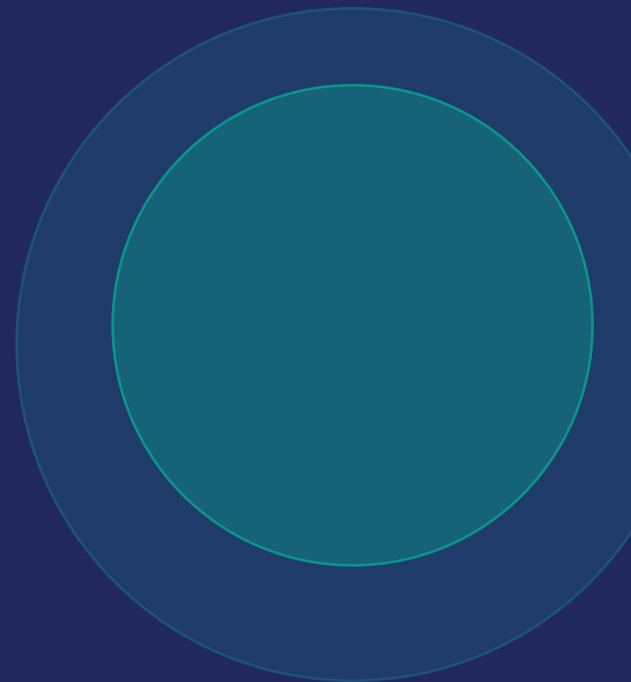


A Q I A N A L Y S I S

Regression & Classification on Indian Monitoring Station Data

Snigdha Dwivedi | Roll No: 2023BTCSE018
Machine Learning (STBT2102) | Dr. Dileep Kumar Singh
Jagran Lakecity University, Bhopal



What is AQI & Why Does It Matter?

Air Quality Index (AQI) is a standardised number (0–500+) that communicates how polluted the air is. Higher AQI = more pollution = greater health risk.

Task 1 — Regression

Predict exact AQI value (0–500+)

"AQI will be 156"

Task 2 — Classification

Predict AQI category (6 buckets)

"Air will be 'Poor'"

AQI Range	Category	Health Impact
0 – 50	Good	Minimal
51 – 100	Satisfactory	Minor
101 – 200	Moderate	Sensitive groups
201 – 300	Poor	Breathing discomfort
301 – 400	Very Poor	Serious effects

Data Source & Features

108,035

Station-day records

26

Cities covered

109

Monitoring stations

2015–2020

Time range

Input Features (Pollutants)

- PM2.5 — Fine particulate matter ($\mu\text{g}/\text{m}^3$)
- PM10 — Coarse particulate matter ($\mu\text{g}/\text{m}^3$)
- NO — Nitric Oxide ($\mu\text{g}/\text{m}^3$)
- NO2 — Nitrogen Dioxide ($\mu\text{g}/\text{m}^3$)
- NOx — Nitrogen Oxides ($\mu\text{g}/\text{m}^3$)
- CO — Carbon Monoxide (mg/m^3)
- SO2 — Sulfur Dioxide ($\mu\text{g}/\text{m}^3$)
- O3 — Ozone ($\mu\text{g}/\text{m}^3$)

Regression Target

AQI (numeric: 0 → 500+)

Central Pollution Control Board (CPCB), India
Kaggle: Air Quality Data in India

Classification Target

AQI_Bucket (6 categories)

Good → Satisfactory → Moderate
Poor → Very Poor → Severe

Cleaning & Feature Engineering

1

Merge Datasets

Joined station_day.csv with stations.csv on StationId → 108,035 rows, 20 columns

Notable Missing Data

Xylene: 78.8% | NH3: 44.5%
PM10: 39.5% | PM2.5: 20.0%

2

Handle Missing Data

Imputed numeric pollutants using each station's own median; dropped Xylene, Benzene, Toluene (>30% null)

3

Drop Null AQI Rows

Removed rows where target (AQI) was missing → 87,025 clean rows retained

4

Rederive AQI_Bucket

Used a lookup function to fill any missing AQI_Bucket from AQI numeric value

5

Scale Features

Applied StandardScaler on all 8 pollutant features before training linear models

Key Insights Before Modeling

Right-Skewed AQI Distribution

Mean AQI = 179.7, Median = 132. Most values cluster in 100–200. Heavy right tail from extreme pollution events.

PM2.5 is the Strongest Predictor

Correlation with AQI = 0.81. Feature importance confirms PM2.5 → PM10 → CO → NO2 across all models.

City-Level Differences

Delhi and Mumbai show highest median AQI. Bengaluru and Chennai are notably cleaner.

Class Imbalance

Moderate (46%) and Satisfactory (27%) dominate. Severe = only 5%. XGBoost handles this better than simple classifiers.

Seasonal & Temporal Patterns

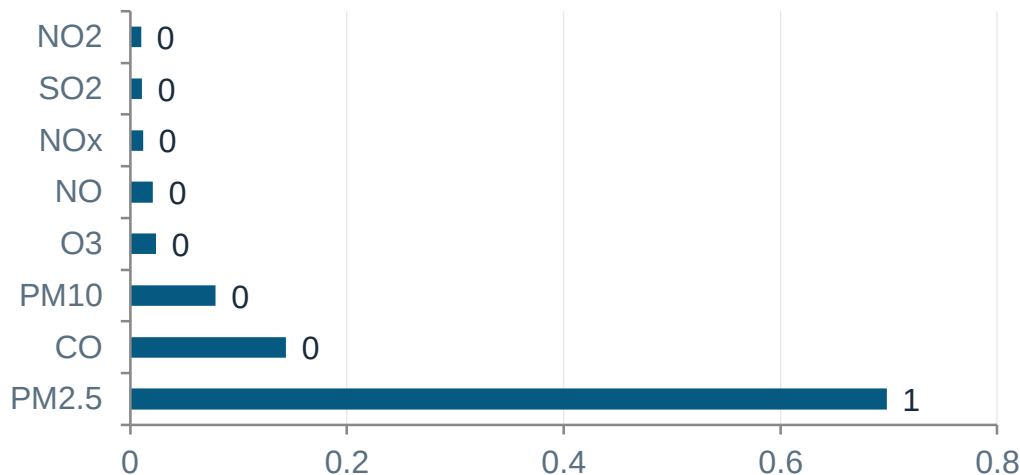
Winter months (Nov–Jan) show significantly higher AQI. Rush hour spikes at 8 AM and 7 PM in hourly data.

Weekend Effect

Weekday average AQI = 168.0 vs Weekend = 164.7. Slight but consistent improvement on weekends.

Predicting Numeric AQI (Task 1)

Random Forest Feature Importance

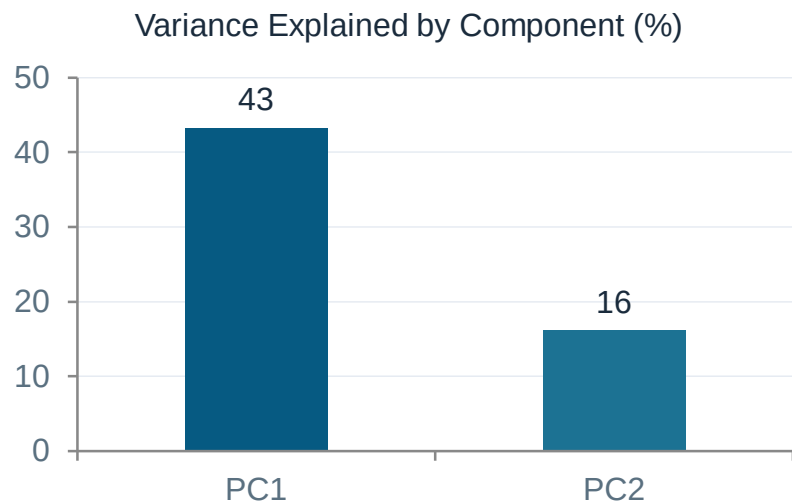


Model	RMSE	R ² Score
Random Forest ★	37.16	0.9200
Linear Regression	53.39	0.8349
Ridge Regression	53.39	0.8349
Lasso Regression	53.39	0.8349

- Random Forest achieved $R^2 = 0.92$ — best by far
- Linear, Ridge & Lasso all plateau at $R^2 = 0.835$
- No overfitting — Ridge/Lasso gave no improvement
- PM2.5 alone contributes 69.8% of RF importance

Best Model: Random Forest | $R^2 = 0.92$ | RMSE = 37.16

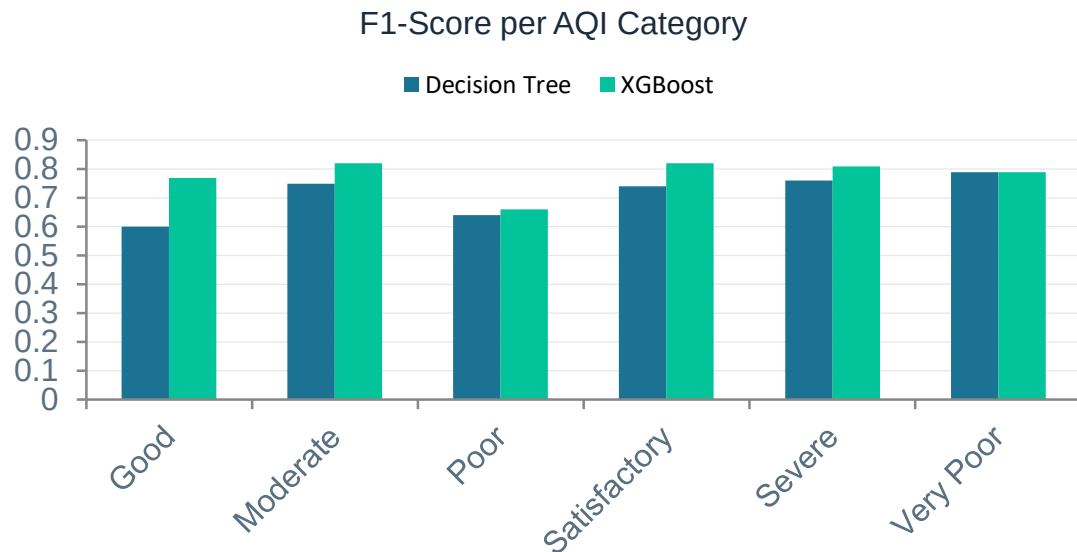
Principal Component Analysis (PCA)



Feature	PC1	PC2
PM2.5	0.397	-0.161
PM10	0.401	-0.196
NO	0.449	-0.165
NO2	0.431	0.066
NOx	0.471	-0.123
CO	0.138	0.660
SO2	0.207	0.633
O3	0.104	0.229

- 2 components capture 59.26% of total variance
- PC1 (~43%): dominated by particulate pollutants — NOx, NO, NO2, PM10, PM2.5
- PC2 (~16%): driven by CO and SO2 — combustion gases
- PCA confirms two distinct pollution sources: particulate vs. combustion

Predicting AQI Category (Task 2)



Model	Accuracy	F1-Score
XGBoost ★	79.06%	0.7902
Decision Tree	73.15%	0.7292

Key Takeaways

- XGBoost outperforms Decision Tree by ~6%
- Class imbalance affects 'Good' & 'Poor' most
- XGBoost handles imbalance better via boosting
- PM2.5 is top feature in both classifiers

Best Classifier: XGBoost | Accuracy = 79.06% | Weighted F1 = 0.79

Final Model Selection & Key Findings

Task 1: Regression

Random Forest

$R^2 = 0.92$ · RMSE = 37.16

Task 2: Classification

XGBoost

Accuracy = 79.06% · F1-Score = 0.79

Key Findings



PM2.5 is the strongest single predictor of AQI across ALL models



Class imbalance exists — 'Severe' = only 5% of data



Weekend AQI is ~3.3 units lower than weekday (cleaner air)



Pollution relationships are non-linear → tree models outperform linear ones



Rush hour pollution spikes at 8 AM and 7 PM in hourly patterns



Ridge & Lasso provided no uplift → no overfitting in linear models